

RPG and LLM



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pre-columbian ceramic llama
Walters Art Museum

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Ollama



[Discord](#) [GitHub](#) [Models](#)

 Search models



**Get up and running with large
language models.**

Run [DeepSeek-R1](#), [Qwen 3](#), [Llama 3.3](#), [Qwen 2.5-VL](#),
[Gemma 3](#), and other models, locally.

Download ↓

Available for macOS,
Linux, and Windows

DMI: Dell Inc. Precision Tower 5810/0K240Y

```
$ lscpu -p
# The following is the parsable format, which can be fed to other
# programs. Each different item in every column has an unique ID
# starting usually from zero.
# CPU,Core,Socket,Node,,L1d,L1i,L2,L3
0,0,0,0,,0,0,0,0
1,1,0,0,,1,1,1,0
2,2,0,0,,2,2,2,0
3,3,0,0,,3,3,3,0
4,4,0,0,,4,4,4,0
5,5,0,0,,5,5,5,0
6,6,0,0,,6,6,6,0
7,7,0,0,,8,8,8,0
8,8,0,0,,9,9,9,0
9,9,0,0,,10,10,10,0
10,10,0,0,,11,11,11,0
11,11,0,0,,12,12,12,0
12,12,0,0,,13,13,13,0
13,13,0,0,,14,14,14,0
14,0,0,0,,0,0,0,0
15,1,0,0,,1,1,1,0
16,2,0,0,,2,2,2,0
17,3,0,0,,3,3,3,0
18,4,0,0,,4,4,4,0
19,5,0,0,,5,5,5,0
20,6,0,0,,6,6,6,0
21,7,0,0,,8,8,8,0
22,8,0,0,,9,9,9,0
23,9,0,0,,10,10,10,0
24,10,0,0,,11,11,11,0
25,11,0,0,,12,12,12,0
26,12,0,0,,13,13,13,0
27,13,0,0,,14,14,14,0
```

```
$ nvidia-smi
Sat Jul 19 10:54:17 2025
+-----+
| NVIDIA-SMI 535.247.01                  Driver Version: 535.247.01   CUDA Version: 12.2   |
+-----+-----+-----+-----+-----+-----+
| GPU  Name      Persistence-M   Bus-Id        Disp.A    Volatile Uncorr. ECC |
| Fan  Temp        Perf          Pwr:Usage/Cap     Memory-Usage | GPU-Util  Compute M. |
|                                           MIG M.       |
+-----+-----+-----+-----+-----+-----+
|   0   NVIDIA GeForce GTX 1650      Off          00000000:02:00:00 Off      N/A
| 27%   29C        P8              5W / 75W           49MiB / 4096MiB      0%      Default
|                                           N/A              |
+-----+-----+-----+-----+-----+-----+

+-----+
| Processes:                               |
| GPU   GI    CI          PID    Type    Process name                        GPU Memory |
|      ID    ID                                   Usage      |
+-----+-----+-----+-----+-----+-----+
|   0   N/A    N/A         2132     G      /usr/lib/xorg/Xorg                  39MiB |
|   0   N/A    N/A         2318     G      /usr/bin/gnome-shell                5MiB |
+-----+-----+-----+-----+-----+-----+

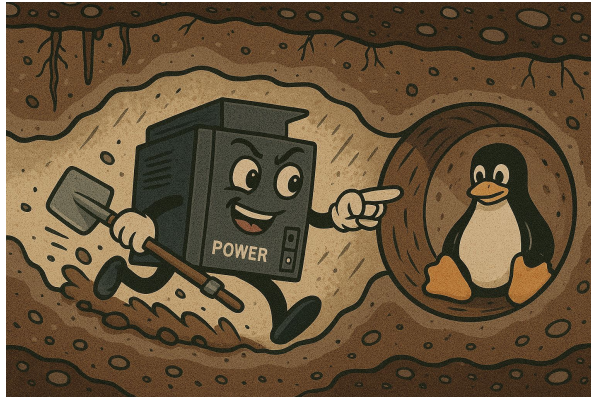
$ lsmem
RANGE                                SIZE  STATE  REMOVABLE  BLOCK
0x0000000000000000-0x000000007fffffff  2G online      yes      0
0x0000000100000000-0x000000014fffffff  80G online     yes    2-41

Memory block size:           2G
Total online memory:         82G
Total offline memory:        0B
```

Reverse tunnel to Linux from IBM i

- Allow ourselves to use Ollama running on the Linux server from IBM i making the port appear local.
 - Necessary because the Linux server does not have a publicly accessible IP address.
- Leave this up as long as we are going to use Ollama from IBM i.

```
ssh -R 11434:localhost:11434 -l myuserprof mysystem.mycompany.com
```



curl the query

```
curl http://localhost:11434/api/chat -d '{
  "model": "llama3.2:3b",
  "stream": false,
  "messages": [
    { "role": "user", "content": "why do dogs bark?" }
  ]
}'
```



Patrick Behr calls ChatGPT from SQL using HTTP_POST

```
CREATE OR REPLACE FUNCTION pbehr.askChatGpt
(
    in_prompt      varchar(1024)
    , in_sysmsg    varchar(512)    default null
    , in_temp      varchar(10)     default '1'
    , in_maxtokens varchar(10)     default '500'
)
RETURNS VARCHAR(4096)

BEGIN

    DECLARE apiUrl  VARCHAR(64)  DEFAULT
'https://api.openai.com/v1/chat/completions';
    DECLARE apiKey  VARCHAR(64)  DEFAULT
'sk-pd38i1R9XAMDVvk34Key4T3BlbkFJuzIV5kX8tKUainKAs4XG';
    DECLARE model   VARCHAR(32)   DEFAULT 'gpt-3.5-turbo';
    DECLARE headers VARCHAR(256)  DEFAULT '';
    DECLARE messages VARCHAR(2500) DEFAULT '';
    DECLARE sysmsg  VARCHAR(1000) DEFAULT '';
```

```
    DECLARE userMsg  VARCHAR(1500) DEFAULT '';
    DECLARE request  VARCHAR(4096) DEFAULT '';

    SET headers = '{"headers":{"Authorization":
"Bearer ' CONCAT apiKey CONCAT '",'
"Content-Type":"application/json"}}';

    IF in_sysmsg IS NOT NULL THEN
        SET sysmsg = '{"role":"system",
"content":"' CONCAT in_sysmsg CONCAT '"},'';
    END IF;

    SET userMsg = '{"role":"user", "content":"'
CONCAT in_prompt CONCAT '"}';

    SET messages = '[' CONCAT sysmsg CONCAT
userMsg CONCAT ']';
```

```
    SET request = '{"model":"' CONCAT
model          CONCAT '",' CONCAT
                '"temperature":"' CONCAT
in_temp        CONCAT '",' CONCAT
                '"max_tokens":"' CONCAT
in_maxtokens   CONCAT '",' CONCAT
                '"messages":"'  CONCAT
messages       CONCAT '}'';

    RETURN (
        SELECT apiResponse
        FROM JSON_TABLE(
            HTTP_POST( apiUrl, request, headers
        ),
            'lax $.choices[0]'
            COLUMNS(
                apiResponse varchar(4096) PATH
                'lax $.message.content'
            )
        )
    );

END;
```

Same sort of thing in Python

```
import ollama
# response = ollama.chat(model='llama3.1:8b', messages=[
# response = ollama.chat(model='granite-code:8b', messages=[
response = ollama.chat(model='llama3.2:3b', messages=[
    {
        'role': 'user',
        'content': 'Why is the sky blue?',
    },
])
print(response['message']['content'])
```



DbToo

- Code: <https://github.com/IBM/AI-SDK-Db2-IBMi/>
- Docs: <https://ibm.github.io/AI-SDK-Db2-IBMi/>

The screenshot shows the documentation page for DbToo, a Db2 for i SDK. The page has a light blue header with the title "A Db2 for i SDK for connecting to various services" and a search bar. A left sidebar contains a navigation menu with links to Home, IBM watsonx, Ollama, OpenAI-compliant endpoints, Kafka, PASE, Slack, SMS (Twilio), and Other Useful links (external). The main content area is titled "DbToo" and "Overview". It includes a welcome message, a note that the project is still in bringup state, and an installation section with a caution about user profile permissions. The right sidebar lists "On this page" links for Overview and Installation.

A Db2 for i SDK for connecting to various services Ctrl K

Home

- IBM watsonx >
- Ollama >
- OpenAI-compliant endpoints >
- Kafka >
- PASE >
- Slack >
- SMS (Twilio) >
- Other Useful links (external) ▾
 - IBM

DbToo

Overview

Welcome! This projects aims to simplify the task of integrating RPG and/or Db2 workloads with external services. The navigation pane on the left lists technologies supported by this experimental SDK.

Note

This project is still in bringup state and should not be deployed to production!

Installation

Caution

The installing user profile **MUST** have the `QIBM_DB_SECADM` function usage authority!

To install on IBM i, you need to clone this `git` repository onto an IBM i. You can do this with the following methods:

1. use `git clone` on this repo
2. download the repo as a .zip from GitHub and extract it to your IBM i

Once the repo sources are in the IFS, then you can use GNU Make to build the SDK. GNU Make,

On this page

- [Overview](#)
- [Overview](#)
- [Installation](#)

Do it in DbToo

```
set
  dbsdk_v1.ollama_protocol = 'http';
set
  dbsdk_v1.ollama_server = 'localhost';
set
  dbsdk_v1.ollama_port = 11434;
set
  dbsdk_v1.ollama_model = 'llama3.2:3b';
select
  dbsdk_v1.ollama_getmodel ()
from
  SYSIBM.SYSDUMMY1;
```

```
select
  dbsdk_v1.ollama_generate (
    prompt => '{ "role": "user", "content": "why do dogs
bark?"}') as myanswer
from
  SYSIBM.SYSDUMMY1;
set
  dbsdk_v1.ollama_model = 'granite-code:8b';
select
  dbsdk_v1.ollama_generate (
    prompt => '{ "role": "user", "content": "why do dogs
bark?"}') as myanswer
from
  SYSIBM.SYSDUMMY1;
```

Do it in SQLRPGLE (v-e-r-y simply)

```
**FREE
// Jack Woehr jack@seidengroup.com 2025-06-09
Ctl-Opt DftActGrp(*No) Option(*Srcstmt : *NodebugIO) main(qOllama);

dcl-s MyAnswer SQLTYPE(clob:16000000);
dcl-s DspStr varchar(52);

dcl-proc qOllama;

    exec sql call dbsdk_v1.ollama_setmodelforme (model => 'granite-code:8b');
    exec sql select dbsdk_v1.ollama_generate (prompt => '{ "role": "user", "content": "why do dogs bark?" }' ) into :MyAnswer
from SYSIBM.SYSDUMMY1;
    DspStr = MyAnswer_DATA;
    dsply DspStr;

    *inlr=*on;
end-Proc;
```

Roo, based on my example

```
**FREE
// Jack Woehr jack@seidengroup.com 2025-09-26
// New RPG program to interact with an Ollama-compatible LLM API
// QSYS/CRTSQLRPGI OBJ(NEWLLM) SRCSTMF('/home/jwoehr/work/AI/COMMONSweden25Q4/code/rpgollama/NEWLLM.SQLRPGLE') TEXT('Roo-authorized take on RPGOLLAMA') CVTCCSID(*JOB)
Ctl-Opt DftActGrp(*No) Option(*Srcstmt : *NodebugIO) main(qNewLLM);

dcl-s MyAnswer SQLTYPE(clob:16000000);
dcl-s DspStr varchar(52);
dcl-s PromptText varchar(200) inz('{ "role": "user", "content": "Tell me a short story about a brave knight."}');

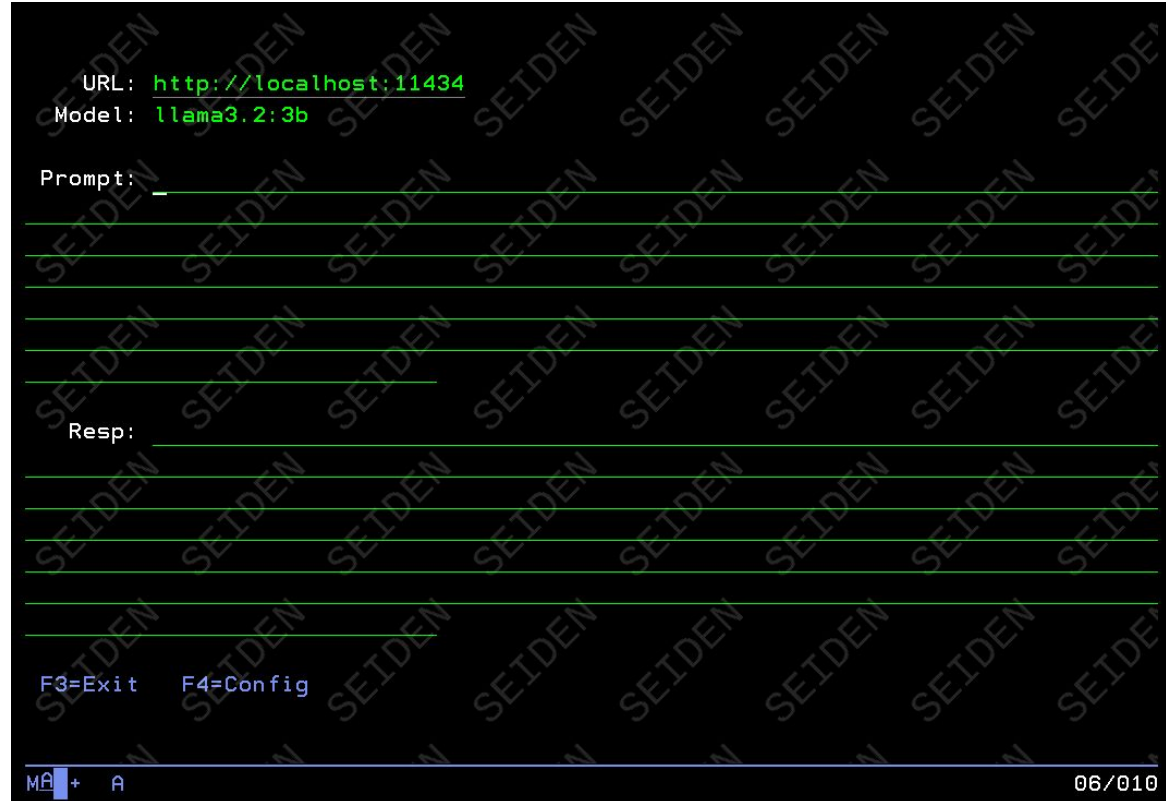
dcl-proc qNewLLM;
// Configure Ollama server settings
exec sql call dbsdk_v1.ollama_setserverforjob(hostname => 'localhost');
exec sql call dbsdk_v1.ollama_setportforjob(port => '11434');
exec sql call dbsdk_v1.ollama_setprotocolforjob(protocol => 'http');
exec sql call dbsdk_v1.ollama_setmodelforjob(model => 'llama3.2:3b'); // Using llama3.2:3b as per user's available model

// Generate response from LLM
exec sql select dbsdk_v1.ollama_generate(prompt => :PromptText) into :MyAnswer from SYSIBM.SYSDUMMY1;
if SQLSTATE = '00000';
    DspStr = MyAnswer_DATA;
    dsply DspStr;
    snd-msg *INFO DspStr %target(*CALLER);
else;
    DspStr = 'SQL Error: ' + SQLSTATE;
    dsply DspStr;
    snd-msg *INFO DspStr %target(*CALLER);
endif;

*inlr=*on;
end-Proc;
```

AI via DSPF

- OLLAMARPG
- GEMQRY



The screenshot shows a terminal window with a black background and green text. At the top, it displays the configuration: URL: <http://localhost:11434> and Model: llama3.2:3b. Below this is a 'Prompt:' label followed by a series of horizontal lines for input. Further down is a 'Resp:' label followed by another series of horizontal lines for output. At the bottom left, it shows 'F3=Exit' and 'F4=Config'. The bottom status bar includes a small icon, a plus sign, the letter 'A', and the page number '06/010' on the right.

```
URL: http://localhost:11434  
Model: llama3.2:3b  
  
Prompt: _____  
_____  
_____  
_____  
_____  
_____  
_____  
_____  
  
Resp: _____  
_____  
_____  
_____  
_____  
_____  
_____  
_____  
  
F3=Exit  F4=Config  
  
MA + A 06/010
```

AI via DSPF

- **OLLAMARPG**
- GEMQRY

Enter the desired values directly or press F4 to prompt.
If the value doesn't fit below, add the value to the config and use the prompt.

Protocol: http

Server: localhost

Port: 11434

Model: llama3.2:3b

F12=Cancel F6=Make My Default Enter=Use these values

MA + A

06/014

AI via DSPF

- OLLAMARPG
- **GEMQRY**

```
// Call the GEMINI_QUERY function
EXEC SQL

SET :hResult = GEMINI_QUERY(
    MODEL => TRIM(:hModel),
    KEY => TRIM(:hKey),
    CONTENT => TRIM(:hContent)
);
```

```
Call Gemini Query w/DTAARA (CALLGEMQYD)

Type choices, press Enter.

Gemini model name . . . . . > 'gemini-2.5-flash'

Query text . . . . . > 'How did dogs evolve?'

F3=Exit  F4=Prompt  F5=Refresh  F12=Cancel  F13=How to use this display
F24=More keys

Bottom

MA + A 05/037
```

Thank you



Any sufficiently advanced technology is indistinguishable from magic. - A. C. Clarke



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